

Ex:-5 VERIFICATION OF NORTON'S THEOREM

Aim:-

To verify Norton's theorem practically and theoretically for the given DC circuit.

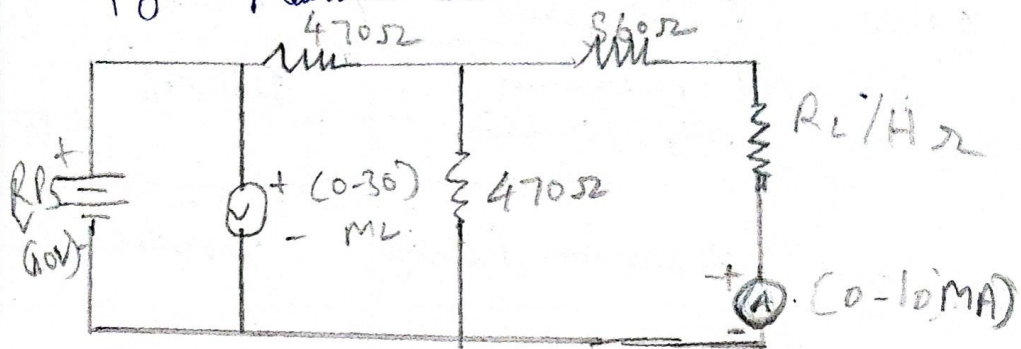
Apparatus Required :-

S.L.No	Apparatus	Specification	Quantity
1.	Regulated Power supply	C 0-30V	1
2.	Voltmeter	C 0-30V) MC	1
3.	Ammeter	C 0-10mA) (MC)	1
4.	Resistor	470 Ω	2
		560 Ω	1
		1k Ω	1
5.	Bread board	- - -	1
6.	Multimeter	- - -	1

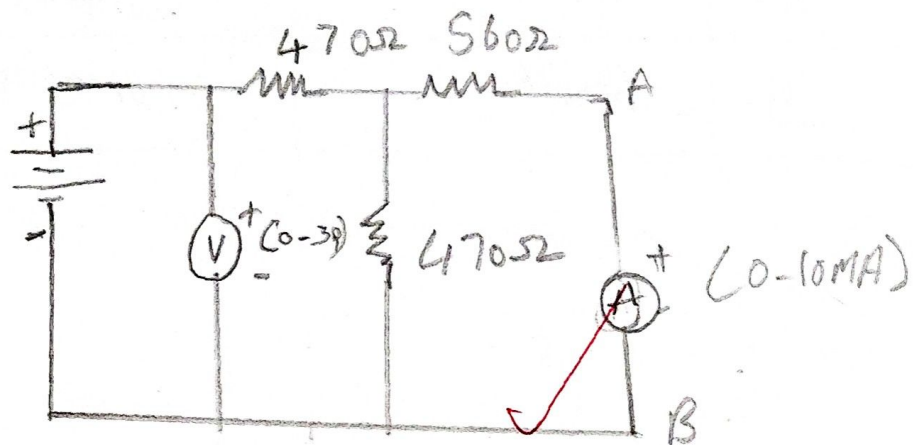
CIRCUIT DIAGRAM:-

THEOREM:-

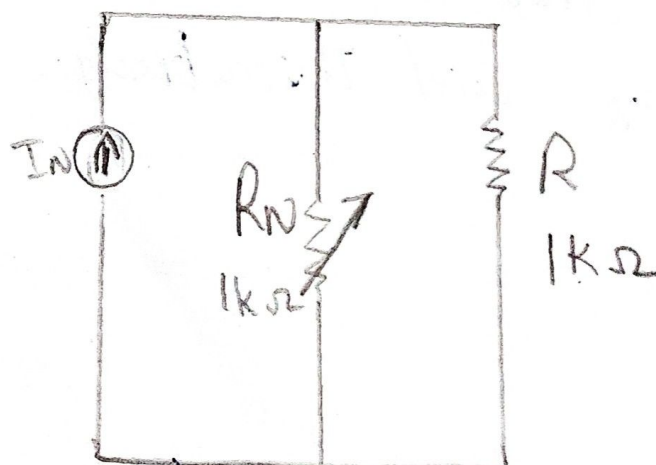
To measure I_L



To measure R_{th} or R_N



Norton's equivalent circuit



Theory:-

NORTON'S THEOREM:-

Norton's Theorem states that, a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a current source, $I_N (= I_{SC})$ in parallel with a resistor $R_N (= R_{TH})$, where $I_N (= I_{SC})$ is the short-circuit current through the load terminal and R_N is the equivalent resistance at the load terminals when the independent source are current off.

Norton's Theorem:-

I_N is the short-circuit current through the load terminal i.e., $I_N = I_{SC}$

Norton's Resistance, R_N :-

R_N is the look back resistance across the load terminal when all the sources are replaced by their internal resistance. An ideal voltage source is replaced by open circuit as its internal resistance is infinity.

TABULAR COLUMN:- 1

To measure I_z

V_1 (volts)	I_z (amps)
10V	275MA

TABULAR COLUMN:- 2

To measure R_{TH} or R_D

V_1 (volts)	R_{TH} (Ω)
10V	774

TABULAR COLUMN:- 3

To measure I_N or I_{sc}

V_1 (volts)	I_N (amps)
10V	6.27MA



PROCEDURE:-

1. Make the connections as per the circuit Diagram - 1
2. Vary the RPS and set an input voltage of 10V.
3. Note down the voltmeter reading (V_i) and ammeter reading (I_3) in Tabular Column 1.
4. Measure the Thevenin's resistance $R_{TH} = \text{Norton's resistance } R_N$.
5. Measure the Thevenin's resistance $R_{TH} = \text{Norton's resistance } R_N$.
6. Switch off the supply and make connections for circuit diagram: 3
7. Set an input voltage of 10V in the RPS and note down the voltmeter readings V_i and $V_{TH} (= V_{OC})$ in Tabular Column: 3
8. Switch off the supply and make connections for circuit Diagram 4.
9. Set an input voltage of 10V in the RPS and note down the voltmeter readings V_i and ammeter reading I_1 and ammeter reading $I_{NV} (= I_{SC})$ in Tabular Column 4.
10. Draw the Thevenin's equivalent circuit and Norton's equivalent circuit as shown in circuit diagram 5 & 6 respectively.
11. Calculate the I_2 value using the formula Thevenin Theorem.

$$I_2 = V_{TH} / (R_{TH} + R_L)$$

Norton's Theorem

$$I_2 = I_N * R_N / (R_N + R_L)$$

12. Theoretically verify the Norton's theorem.

Calculation:-

Theoretical Calculations

short circuit the load resistance

Measure I_N :-

Parallel

$$R_{eq1} = 255.5 \Omega$$

$$R_{eq2} = 235 \Omega$$

$$\begin{aligned} \text{Total} &= 255.5 + 470 \Omega \\ &= 725.5 \Omega \end{aligned}$$

$$I_s = \frac{V}{R_T} = \frac{10}{255.5}$$

$$= 0.013 A$$

$$I_N = 0.013 \times \frac{470}{470 + 560}$$

$$I_N = 6.75 \text{ mA}$$

To measure R_N :-

$$R_{eq} = 560 + 235$$

$$R_N = R_2 + \frac{R_1 + R_3}{R_1 + R_3}$$

$$\begin{aligned} &= 560 + 235 \\ &= 795 \Omega \end{aligned}$$

Load current

$$I_L = I_N \frac{R_N}{R_N + R_L}$$

$$= 6.75 + \frac{795}{795 + 1000}$$

$$= 2.768 \text{ mA}$$

$$V_2 = 2.768V$$

Marks SPLIT - UP

Sl. No.	Description	Mark allotted	Mark obtained
1.	Experimental setup / connections	20 marks	18
2.	Observation / Reading	20 Marks	18
3.	Calculation / Results	20 marks	18
4.	VIVA	20 marks	18
5.	Record	20 marks	18

90

Result :-

Thus Thevenin's and Norton's Theorem is
 Verified Practically and Theoretically.

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